Multimodal data and fusion

Multimodal data

Medical image



pixels

Radiology report

FINDINGS
Lungs are clear. Heart size is normal. No effusion or pneumothorax.

IMPRESSION
No acute cardiopulmonary disease.

free text

Labs and demographics

Age	54
Sex	F
WBC	11.2 $\times 10^9$/L
CRP	24 mg/L
Hb	13.1 g/dL

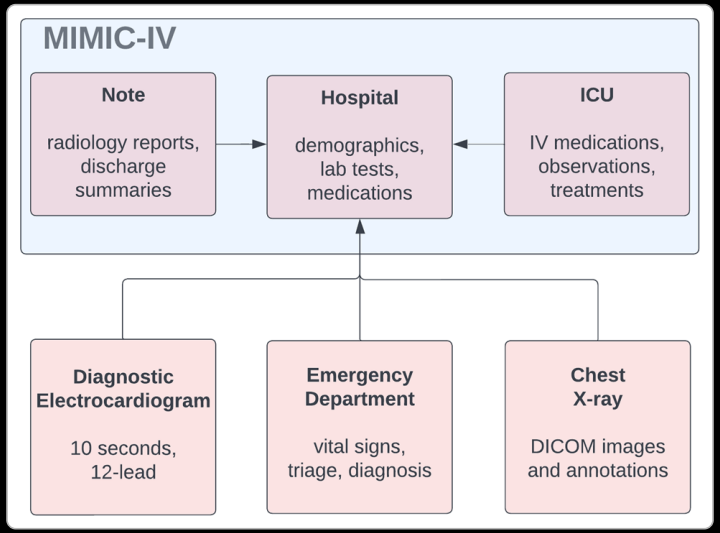
numbers and categories

one patient

Biobanks, health records, imaging and wearable sensors now set the stage for **multimodal AI** that captures the **complexity of human health and disease**.

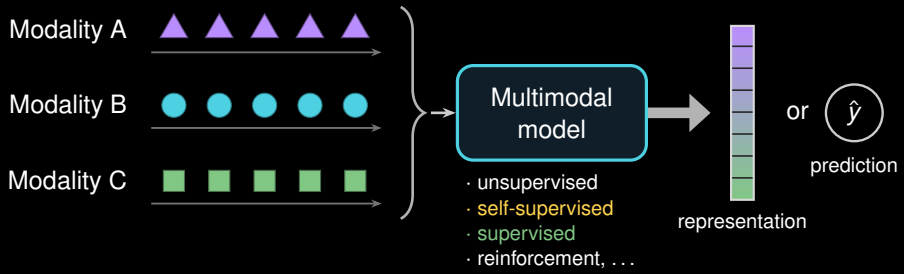
Acosta et al., *Multimodal biomedical AI*, Nature Medicine

MIMIC-IV, A FREELY ACCESSIBLE ELECTRONIC HEALTH RECORD DATASET



Johnson et al., *MIMIC-IV, a freely accessible electronic health record dataset*, Scientific Data

MULTIMODAL CLINICAL AI

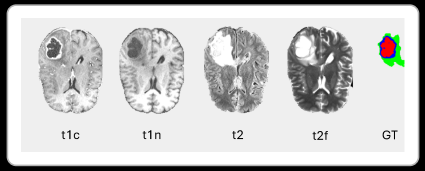


Morency & Liang, *Multimodal Machine Learning*, CMU 11-777

More on multimodal data

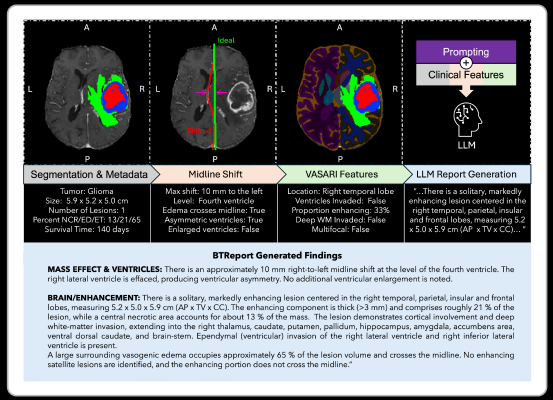
Special case

Nearly identical MRIs from tumor segmentation tasks.



Common case

MRI and radiology report.

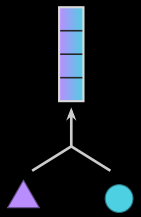


Ren et al., *Here Comes the Explanation: A Shapley Perspective on Multi-contrast Medical Image Segmentation*, arXiv; Heras Rivera et al., *BTReport: A Framework for Brain Tumor Radiology Report Generation with Clinically Relevant Features*, MIDL

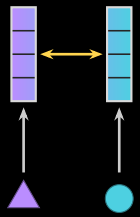
THREE DIFFERENT STRATEGIES IN MULTIMODAL LEARNING

Representation: learning representations that reflect cross-modal interactions between individual elements, across different modalities.

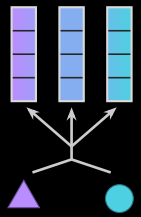
Fusion



Coordination

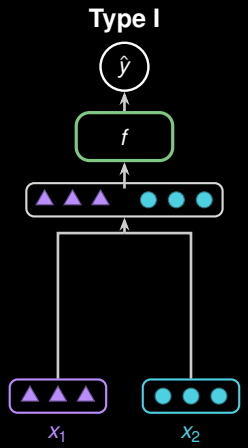


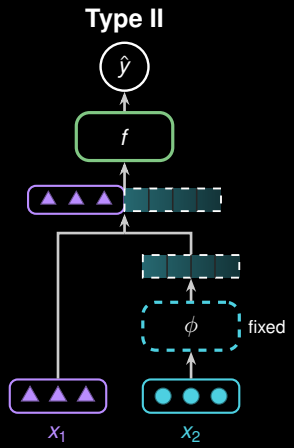
Fission



Early, joint and late fusion

EARLY FUSION





$\hat{y} = f([x_1 ; x_2])$ $\min_f \mathcal{L}$

Original inputs, concatenated.

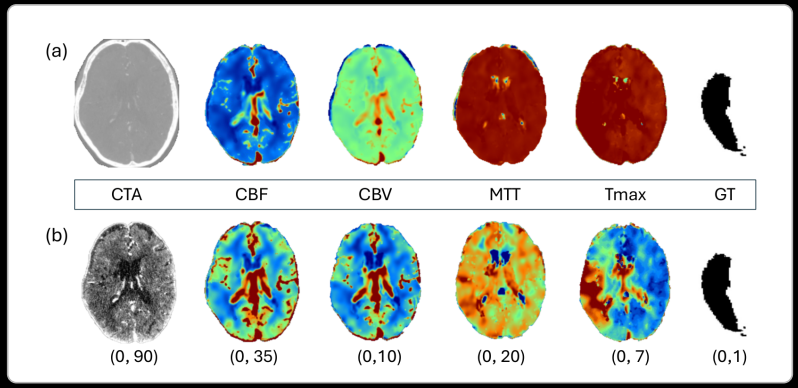
$\hat{y} = f([x_1 ; \phi(x_2)])$ $\min_f \mathcal{L}$

One input pre-extracted by a fixed ϕ.

Fuse before the model; the loss trains only f.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

Early fusion: stack the modalities as channels



CTA CT angiogram: the blood vessels

CBF cerebral blood flow: how fast blood reaches the tissue

CBV cerebral blood volume: how much blood is in the tissue

MTT mean transit time: how long blood takes to pass through

Tmax time to maximum: how late the contrast arrives

GT ground truth: the lesion, traced on MRI days later

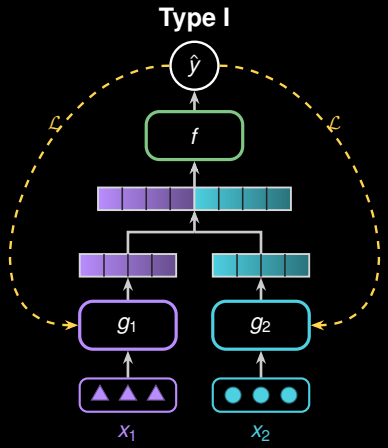
Row (a) is each map as acquired; row (b) is the same map after windowing.

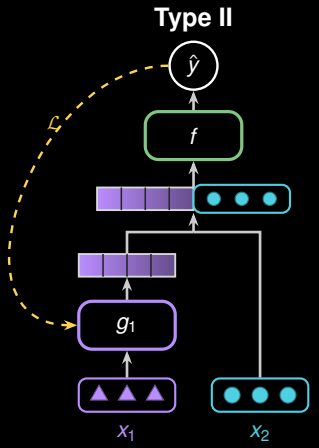
Heras Rivera et al., *Clinically-Informed Preprocessing Improves Stroke Segmentation in Low-Resource Settings*, MICCAI MIRASOL Workshop

Limitations of input-level stacking

- **Incompatible spatial data.** A 3D MRI volume and a 2D X-ray projection differ in dimensionality and spatial correspondence, preventing direct channel-wise stacking.
- **Heterogeneous data types.** Clinical notes, laboratory measurements, and medication records lack the spatial structure required for direct stacking with image inputs.
- **Missing modalities at inference.** Models trained on a fixed set of input channels generally require all corresponding modalities at inference.
- **Alternative: joint fusion.** Modality-specific encoders map each input to a feature representation, which can then be combined for downstream prediction.

JOINT FUSION





$$\hat{y} = f([\,g_1(x_1)\,;\,g_2(x_2)\,]) \quad \min_{f,\,g_1,\,g_2} \mathcal{L}$$

Every modality gets a trained encoder.

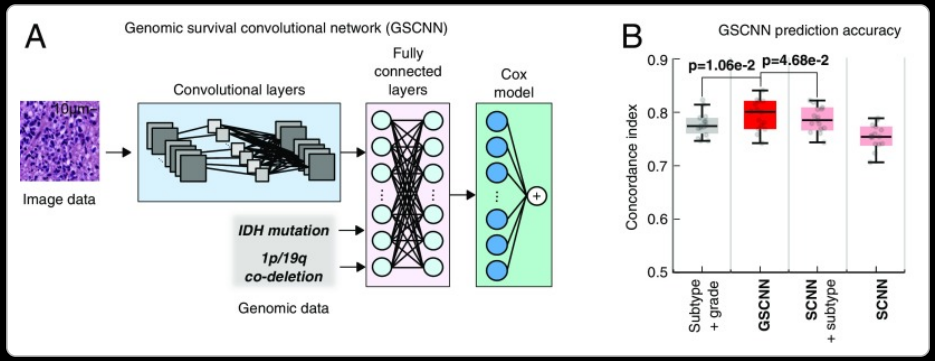
$$\hat{y} = f([\,g_1(x_1)\,;\,x_2\,]) \quad \min_{f,\,g_1} \mathcal{L}$$

One encoder; the other input joins raw.

Fuse learned **features**; the loss reaches every encoder.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

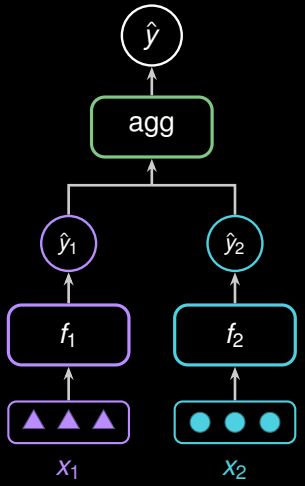
JOINT FUSION: HISTOLOGY AND GENOMICS



For glioma survival, IDH and 1p/19q status enter the CNN's fully connected layers and one Cox loss trains the whole network (**joint fusion, Type II**), raising the median c-index from 0.754 (histology alone) to **0.801**.

Mobadersany et al., *Predicting cancer outcomes from histology and genomics using convolutional networks*, PNAS

Late fusion



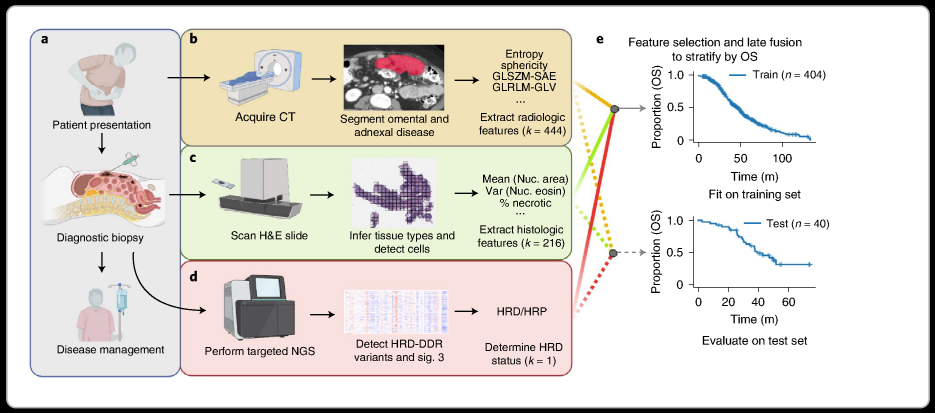
$$\hat{y} = \mathrm{agg}\big(f_1(x_1), f_2(x_2)\big)$$

$$\min_{f_1} \mathcal{L}(\hat{y}_1, y) \qquad \min_{f_2} \mathcal{L}(\hat{y}_2, y)$$

Fuse **predictions**. Each modality is its own model, trained alone; agg is a fixed rule such as an average, or a small trained model.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

LATE FUSION: CT, HISTOLOGY AND GENOMICS

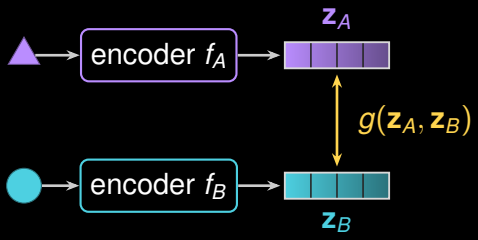


For ovarian cancer survival, each modality gets its own Cox model and a final Cox model fuses their risk scores. The best test model used **CT and histology alone** (c-index 0.62); adding all four modalities did **worse**. More modalities are no guarantee.

Boehm et al., *Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer*, Nature Cancer

Coordination

COORDINATION



Learning with a coordination function:

$$\mathcal{L} = g\big(f_A(\blacktriangle),\ f_B(\bullet)\big)$$

Trained jointly over θ_g, θ_{f_A} and θ_{f_B}.

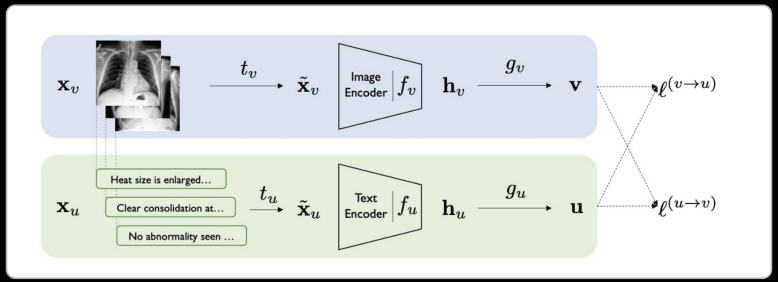
Requires paired data: every triangle needs its own circle.

Example: cosine similarity

$$g(\mathbf{z}_A, \mathbf{z}_B) = \frac{\mathbf{z}_A \cdot \mathbf{z}_B}{\|\mathbf{z}_A\| \, \|\mathbf{z}_B\|}$$

1 when the two embeddings are parallel in latent space

ConVIRT: contrastive learning representations of images and text pairs



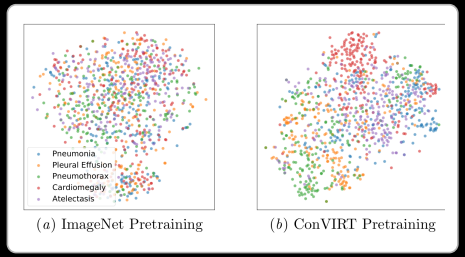
217k image and text pairs from MIMIC-CXR. A **ResNet-50** encodes the image, a **BERT** text encoder initialized from ClinicalBERT encodes the report, each with its own projection head.

$$\ell_i^{(v\to u)} = -\log \frac{\exp\left(\langle v_i, u_i\rangle/\tau\right)}{\sum_{k=1}^{N}\exp\left(\langle v_i, u_k\rangle/\tau\right)} \qquad \mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left(\lambda\,\ell_i^{(v\to u)} + (1-\lambda)\,\ell_i^{(u\to v)}\right)$$

$\langle\cdot,\cdot\rangle$ is cosine similarity and τ a temperature; $\ell_i^{(u\to v)}$ is the same with image and text swapped.

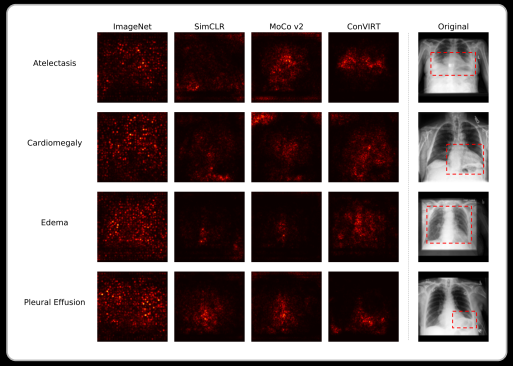
Zhang et al., *Contrastive Learning of Medical Visual Representations from Paired Images and Text*, MLHC

ConVIRT: results



(a) ImageNet Pretraining (b) ConVIRT Pretraining

t-SNE of image embeddings for five conditions. After ConVIRT pretraining (b), images of the same condition cluster more clearly than after ImageNet pretraining (a).



Saliency maps for four abnormalities. ConVIRT's maps match the regions an expert marked (red boxes) most closely.

Zhang et al., *Contrastive Learning of Medical Visual Representations from Paired Images and Text*, MLHC